01 / Purpose, scope and questions

GRAVITY PILOT - 24 VIDEOS | Review protocol v1.0 | 9 October 2026

This document specifies a small test of predictive compression in frozen JEPA features. It is a proposed protocol, not a result. Only document preparation is authorized in this task. A later coding agent must stop after making the 24 videos available for inspection and wait for explicit video approval before extracting embeddings or fitting readers.

Questions being investigated

Q1. Can a linear reader of the full final-layer JEPA embedding predict a ball's position at four seconds using only the first two seconds of video?

Q2. Does that prediction generalize to an initial position excluded from reader training?

Q3. Does a single fixed 16-dimensional projection preserve future-position accuracy within a prespecified 10% relative RMSE tolerance?

Q4. Is orange-versus-blue ball color readable from the full and projected embeddings, and does projection reduce that readability?

Q5. How do paired color changes affect position predictions when the true future position is unchanged?

Q6. What feature-storage reduction is achieved, and what happens in each of the six held-out position groups?

Permitted conclusion

The pilot measures accessibility of future position and color under specified linear readers, plus transfer across six starting positions. It does not estimate Shannon information loss, a universal abstraction-depth score, meta-learning, unseen-gravity generalization or understanding of new object classes. Any detail loss caused by the added projection must not be attributed to JEPA itself.

Exactly 24 clips; one pretrained checkpoint; one encoder layer; one projection seed; no model training, layer sweep, PCA, alternative projection dimensions, new gravity values or search for a favorable result. The older 100-video documents remain a separate historical plan.

02 / Dataset and deterministic scene plan

Let s index six initial positions, a index two gravities and c index two colors. Each scene has all four gravity-color combinations. Every clip contains one ball.

$$s \in \{0,1,2,3,4,5\}, \quad a,c \in \{0,1\}, \quad N = 6 \cdot 2 \cdot 2 = 24$$

$$g_0 = 9.8, \quad g_1 = 4.9 \quad [\mathrm{m} / \mathrm{s}^2]$$

c=0 is orange RGB (235,100,60); c=1 is blue RGB (60,140,235). Use identical colors for both gravities. Background RGB (32,32,32); static grid RGB (44,44,44). Color change includes brightness change; it is not a luminance-matched control.

$$s = 3j + i, \quad i \in \{0,1,2\},\ j \in \{0,1\}$$

$$x_{0,s} = -40 + 40i, \quad y_{0,s} = 140 + 20j$$

$$v_x = 2\ \mathrm{m} / \mathrm{s}, \quad v_y = 0, \quad r = 5\ m$$

World: x in [-100,100] m and y in [0,200] m. Fixed orthographic camera. The six (x0,y0) pairs are (-40,140), (0,140), (40,140), (-40,160), (0,160), (40,160). Only starting position varies across scene groups. No collisions, friction, shadows, textures, occluders or camera motion.

Render 384 x 384 RGB at 30 FPS with 151 frames. Source timestamps are k/30 seconds, k=0,...,150. This provides five actual simulated seconds and 151/30 = 5.0333 seconds of playback. Do not loop, slow or pad a shorter trajectory.

Use lossless FFV1 MKV with decoded RGB equality checked against rendered frames. Filenames: s00_g98_orange.mkv, s00_g98_blue.mkv, s00_g49_orange.mkv, s00_g49_blue.mkv; repeat through s05. Exactly 24 files are experimental clips.

03 / Physics, image mapping and hidden target

$$x_{s,a}(t) = x_{0,s} + 2t, \qquad y_{s,a}(t) = y_{0,s} - \frac{1}{2} g_a t^2$$

These equations determine the rendered trajectory and simulator ground truth. Evaluate them in float64. Coordinates increase rightward and upward in the physical world.

$$u = 1.92(x + 100), \qquad v = 1.92(200 - y), \qquad r_{\mathrm{px}} = 9.6$$

Draw grid lines at pixel coordinates 0,38,76,...,380, then draw the ball with a filled Pillow ellipse bounding box (u-9.6,v-9.6,u+9.6,v+9.6), without added antialiasing. Pin and record the Pillow version. Throughout 0 to 5 seconds, centers lie in x=[-40,50], y=[17.5,160], so radius-five balls remain visible.

Observed input and prediction target

$$k_l = 4l, \quad t_l = \frac{2l}{15}, \quad l = 0, \ldots, 15$$

Only source frame indices [0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60] enter the encoder. They cover t=0 to t=2 seconds. Frames 61 through 150 must never enter the encoder, preprocessing statistics or representation selection.

$$p_{s,a} = (x_{0,s} + 8,\ y_{0,s} - 8g_a)^{\top} \quad [m]$$

The target is center position at t=4 seconds (source frame 120). The formula above is evaluated by the simulator; it is not supplied as a rule to the learned reader. Future-position labels of training groups are allowed for reader fitting. Labels of the held-out group are used only for scoring.

Color is not an input to the position reader. Gravity, initial position, filename, scene ID, timestamps as extra features and simulator parameters are never concatenated to embeddings. Metadata is used only for construction, splitting, validation and baselines.

04 / Frozen representation and projection

Proposed model: V-JEPA 2.1 ViT-B/16 EMA encoder, official source revision 204698b45b3712590f06245fbfba32d3be539812; checkpoint vjepa2_1_vitb_dist_vitG_384.pt. Verify the archived source and strict EMA weight loading in the eventual environment. Never substitute random weights, another checkpoint or another feature layer. Do not run inference during document preparation.

$$\widetilde{I}_{l,h,w,c} = \frac{I_{l,h,w,c} / 255 - \mu_c}{\sigma_c}$$

RGB means (0.485,0.456,0.406), standard deviations (0.229,0.224,0.225). Input [1,3,16,384,384]; float32 evaluation; frozen parameters; no crop, resize, augmentation or predictor features. Final post-LayerNorm encoder tokens with hierarchical output disabled.

$$T \in \mathbb{R}^{4608 \times 768}, \qquad z = \frac{1}{4608} \sum_{n=1}^{4608} T_{n,:}$$

The expected 4608 tokens correspond to 8 temporal positions times 24 times 24 spatial positions. Assert this shape; an unexpected shape is a blocker, not permission to pool a different output. Save one raw, unnormalized float32 768-vector per clip.

$$R \in \mathbb{R}^{16 \times 768}, \qquad R_{ij} = \frac{\xi_{ij}}{\sqrt{16}}, \quad \xi_{ij} \in \{-1, +1\}$$

$$u = Rz \in \mathbb{R}^{16}$$

Generate R once using NumPy Generator(PCG64(20261009)); draw integers 0 or 1 in row-major shape (16,768), convert by 2*b-1, then divide by 4. Save R as float32 and record its SHA-256. Do not choose the seed using results. Projection and accumulation use float32; reader calculations then cast saved features to float64.

This is a single data-independent random projection, not PCA or an optimized abstraction. Compare full z against projected u. No L2 normalization before pooling or projection.

05 / Six folds and training-only scaling

$$\mathcal{U}_f = \{(s,a,c): s = f\}, \quad \mathcal{T}_f = \{(s,a,c): s \neq f\}$$

$$f = 0, \ldots, 5, \qquad |\,\mathcal{T}_f\,| = 20, \quad |\,\mathcal{U}_f\,| = 4$$

Leave one starting position out. All four versions of that position remain together. Each video receives exactly one out-of-fold prediction. Fit separate readers for each fold and representation. The same 24 frozen embeddings are reused; six folds do not require six inference runs.

For either raw representation r=z or r=u, compute the feature mean and a single scalar RMS scale from the 20 training vectors only:

$$\bar{r}_f = \frac{1}{20} \sum_{i \in \mathcal{T}_f} r_i$$

$$s_f = \sqrt{\frac{1}{20} \sum_{i \in \mathcal{T}_f} \|r_i - \bar{r}_f\|_2^2}$$

$$a_i = \frac{r_i - \bar{r}_f}{s_f}$$

Apply the same mean and scale to training and held-out features. No per-channel standardization: the scalar scale makes total training feature variance equal to one and fixes the regularization convention across dimensions. If s_f <= 1e-12, stop that representation/fold as degenerate and report it; do not silently replace the scale.

Fit no hyperparameters on the six test groups. Use the fixed ridge penalty lambda=0.01 for both representations and both readers. No nested model selection or alternative penalties. Save training means, scales, coefficients and split IDs.

This is a very small feasibility study: 24 clips but only six scene groups. Reused training folds are dependent. Report all six held-out groups; do not treat clips or folds as independent trials for a significance claim.

06 / Linear readers: complete fitting rule

Use the same ridge least-squares solver for position regression and color classification. Let A contain 20 training feature rows a_i, and let Y contain the corresponding target rows. Position targets are p_i/100 (two columns). Color targets are -1 for orange and +1 for blue (one column). The 100-metre output scale is fixed, not estimated from data.

$$\bar{Y} = \frac{1}{n}\sum_i Y_i, \quad Y_c = Y - 1\bar{Y}, \quad n = 20$$

$$\hat{W} = \arg\min_W \left\{ \frac{1}{n}\|AW - Y_c\|_F^2 + \lambda\|W\|_F^2 \right\}$$

$$\hat{W} = A^\top (AA^\top + n\lambda I_n)^{-1} Y_c, \qquad \lambda = 0.01$$

Implement a linear solve, not an explicit inverse. The dual form is a 20 x 20 system even for 768-dimensional embeddings. Because features are centered, adding the training target mean implements an unpenalized intercept.

$$\hat{Y}_i = a_i \hat{W} + \bar{Y}$$

$$\hat{p}_i = 100\hat{Y}_i \quad \text{(position reader)}$$

$$\hat{c}_i = 1[\hat{Y}_i > 0] \quad \text{(color reader)}$$

A zero color score maps to orange. Do not clip scores or interpret them as probabilities: this is a ridge classifier, not logistic regression. Do not report cross-entropy, bits of mutual information or calibrated uncertainty from these scores.

Training contains ten orange and ten blue clips in every fold. The chance/reference balanced accuracy is 0.5. Position and color readers are separate fits; color-reader weights must not influence the position reader or projection.

07 / Position metrics and motion baseline

$$e_i^{(r)} = \|\hat{p}_i^{(r)} - p_i\|_2$$

$$E_r = \sqrt{\frac{1}{24}\sum_{i=1}^{24} (e_i^{(r)})^2} \quad [m]$$

r is full, projected or baseline. E_r is Euclidean position RMSE, not a per-coordinate RMSE. Also report x- and y-coordinate RMSE, six group RMSEs (four clips each), both gravity-specific RMSEs and all 24 errors. Do not average group RMSEs and label it overall RMSE.

$$t_a = \frac{56}{30}, \quad t_b = 2, \quad \Delta = t_b - t_a = \frac{2}{15}$$

$$\hat{v} = \frac{p(t_b) - p(t_a)}{\Delta}$$

$$\hat{p}_{\mathrm{CV}}(4) = p(t_b) + 2\hat{v}$$

Primary motion baseline: constant-velocity extrapolation using exact simulator centers at the last two observed encoder timestamps. It is deliberately given clean past-position measurements; it receives no future positions or gravity value. Label it an oracle-observation constant-velocity baseline, not a raw-pixel tracking algorithm.

$$e_{\mathrm{CV}}(g) = \frac{32}{15}g$$

Analytical check: horizontal error is zero; downward acceleration makes the baseline predict too high. Error is 20.9066667 m at g=9.8 and 10.4533333 m at g=4.9. Their balanced aggregate RMSE is sqrt((20.9066667^2+10.4533333^2)/2). These are protocol checks, not measured JEPA results.

$$G_r = 1 - \frac{E_r}{E_{\mathrm{CV}}}, \qquad \rho = \frac{E_{\mathrm{projected}}}{E_{\mathrm{full}}}$$

G_r > 0 means better position prediction than this baseline. rho <= 1.10 is the fixed compression-retention criterion. Report absolute errors too: relative retention alone can preserve a poor predictor. If E_full <= 1e-9 m, mark rho undefined and report the absolute change instead; do not improvise a replacement pass rule.

07B / Training-prior baseline and shortcut check

Training-prior baseline

$$\bar{d}_f = \frac{1}{20} \sum_{i \in \mathcal{T}_f} (p_i(4) - p_i(0))$$

$$\hat{p}_{\text{prior, i}} = p_i(0) + \bar{d}_f$$

This baseline receives the exact initial center and learns only average displacement from training labels. It has no query motion or gravity input. Its error is 19.6 m for either gravity in this balanced design. Report both baselines and require both full and projected features to beat both: predicting average gravity must not count as useful motion extraction.

Why this control is necessary

A reader can exploit the fixed velocity and the average training fall distance without distinguishing the two gravity conditions. The two baselines provide different comparisons: recent observed motion versus initial position and average training displacement. The training-prior control checks that a learned reader does more than predict an average fall. It need not outperform constant velocity to be informative.

$$\bar{d}_f = (8, -58.8)^{\top} \quad [m]$$

$$E_{\text{prior}} = 19.6\, m$$

Every fold contains equal numbers of both gravities, so the training average is the same in all folds. At four seconds the two true vertical displacements are -78.4 m and -39.2 m. Their mean is -58.8 m, giving absolute vertical error 19.6 m for each. Horizontal prediction is exact. These values are analytical validation checks, not experimental model results.

Both baselines use exact observed simulator centers and must be labeled accordingly. The prior baseline uses only t=0 from a held-out clip; the constant-velocity baseline uses t=56/30 and t=2. Neither reads the hidden held-out continuation. Record separate errors for each baseline in predictions.csv and summary.json.

Reader comparison

The full and projected learned readers must each beat the smaller of the two baseline RMSEs. Keep the 10% projected-versus-full tolerance as an additional condition. A result that only beats constant velocity is insufficient for the primary claim.

08 / Color, pairing and storage calculations

$$A_r = \frac{1}{24}\sum_i 1[\hat{c}_i^{(r)} = c_i], \qquad \Delta A = A_{\text{full}} - A_{\text{projected}}$$

Report total correct/24, accuracy, per-color recall, confusion matrix and six group accuracies. With balanced held-out colors, balanced accuracy equals ordinary accuracy. A decrease of at least 3/24 = 0.125 is the prespecified descriptive marker of reduced color readability, provided A_full > 0.5. It is not a significance threshold. Near-chance accuracy does not prove color information is absent.

Paired color sensitivity of position predictions

$$J_r = \sqrt{\frac{1}{12}\sum_{s=0}^{5}\sum_{a=0}^{1}\left\|\hat{p}_{s,a,1}^{(r)} - \hat{p}_{s,a,0}^{(r)}\right\|_2^2}$$

The paired videos have identical physical trajectories, so true target difference is zero. J_r measures sensitivity of the learned position prediction to the chosen color change. Report J_full and J_projected descriptively, alongside accuracy. A constant, wrong predictor also has J=0, so this cannot establish useful invariance by itself.

$$b_{\text{full}} = 768 \cdot 32 = 24576 \text{ bits} = 3072 \text{ bytes}$$

$$b_{\text{projected}} = 16 \cdot 32 = 512 \text{ bits} = 64 \text{ bytes}$$

$$\frac{b_{\text{full}}}{b_{\text{projected}}} = 48, \qquad 1 - \frac{16}{768} = 97.9167\%$$

This is a 48-fold reduction of per-example feature payload at float32 precision, not 97.9% loss of Shannon information. For 24 examples the raw payloads are 73,728 and 1,536 bytes, excluding file headers.

Account separately for fixed overhead: a stored float32 projection matrix costs 16 x 768 x 4 = 49,152 bytes. Including that matrix, the projected dataset payload is 50,688 bytes, not 1,536 bytes. The seed recipe can replace matrix storage for regeneration if the numerical implementation is pinned. Encoder, reader and provenance storage are not included in feature-only ratios.

09 / Decision rules and limited breadth

Use the prespecified rules below without choosing different targets, groups or seeds after examining outcomes. If required outputs are invalid, report an incomplete or unevaluable result rather than dropping unfavorable examples.

Primary prediction-retention result

$$E_{\text{full}}, E_{\text{projected}} < \min\left(E_{\text{CV}}, E_{\text{prior}}\right), \quad \rho \leq 1.10$$

All three must hold for the narrow statement: the chosen 16-D projection retained useful position prediction within the chosen tolerance on these held-out groups. This is a descriptive pilot result, not statistical noninferiority or proof of optimal compression.

Additional selective-accessibility pattern

$$A_{\text{full}} > 0.5, \qquad \Delta A \geq 0.125$$

If these also hold, report that future-position prediction was retained while color became less readable by the specified ridge classifier. If both color and position remain readable, report compact accessibility without selective color loss. If full features predict poorly, do not conclude that compression is effective.

Narrow breadth across starting positions

$$b_r = \frac{1}{6}\sum_{f=0}^{5} 1\left[E_{r,f} < \min\left(E_{\text{CV},f}, E_{\text{prior},f}\right)\right]$$

Report this only as fraction of held-out position groups beating both baselines, together with all six RMSEs. It is not world breadth: readers are retrained across folds; both gravities and both colors occur in training; velocities, shapes and physical law families are fixed. No universal depth score is computed.

Also tabulate the retention ratio in each fold where defined. Avoid pooled p-values and independent-binomial confidence intervals: there are only six scene groups, two conditions per factor, and folds share training data. A mixed or null result is a valid outcome.

No additional hypothesis is introduced by showing paired sensitivity or per-coordinate errors; these are declared diagnostics. Do not choose the final narrative using whichever statistic looks most favorable.

10 / Coding-agent brief: staged execution

MISSION: implement this exact 24-video pilot after the user selects the execution environment. Read protocol, pilot_config.json and scene_plan.csv first. Preserve the separate 100-video plan and all synced sources. Do not run the older run.py as if it implements this pilot.

Stage A - implementation and stimulus review

1. Record environment, source revision and dependency versions. Build separate generate, validate, extract, analyze and report commands. Implement the specified formulas with no scientific defaults left implicit.

2. Run software-only checks on hand-computable vectors and analytic trajectories. Generate exactly 24 five-second clips. Validate count, timestamps, visibility, lossless RGB round-trip, identical color-pair trajectories and matching first frames within same-color gravity pairs.

3. Deliver a local gallery listing all 24 clips with scene, gravity and color labels, plus a contact sheet. Ensure every clip can be opened individually. An optional viewing transcode must be labeled derivative and must not enter inference.

4. STOP. Ask the user to inspect and approve the clips. Do not load the checkpoint for inference, extract embeddings, run readers or report empirical model results before this explicit approval. Record the approval with the dataset-manifest hash.

Stage B - only after dataset approval

5. Strictly load the specified encoder and extract 24 early-window embeddings. Masking future frames after full-video encoding is forbidden. The encoder must never receive frames after t=2 seconds.

6. Save the one seeded projection, compute projected embeddings, then fit the 24 readers: 6 folds x 2 representations x 2 tasks. Use training-only scaling, a dual ridge solve and the fixed penalty.

7. Produce every declared metric, table and plot. Report error in metres and storage separately. Do not rerun with a different seed or penalty because the result is poor. A defect correction requires a documented change and rerun of affected outputs.

8. Deliver the reproducible bundle and stop. Do not add gravity values, projection sizes, additional layers, nonlinear readers, significance claims or additional scenes without a new instruction.

11 / Required artifacts and acceptance checks

Deliverables

videos/: 24 verified FFV1 clips. gallery.html: all 24 clips with open/download links and contact sheet. manifest.json: IDs, scene parameters, times, colors, codec and SHA-256. dataset_validation.json: explicit checks and failures.

embeddings_full.npy [24,768], embeddings_projected.npy [24,16], projection.npy [16,768], embedding_index.csv, fitted_readers/ with all fold means/scales/weights. provenance.json: exact source, checkpoint hash, dependency versions, device, precision and input frame indices.

predictions.csv: fold, scene, gravity, color, true x/y at 4 s, full/projected/CV/prior predicted x/y, errors, color scores and predicted labels. summary.json: all pooled/group metrics and decision flags; invalid values represented as null with a reason.

Plots: six-group position RMSE comparison; full/projected color correct counts; paired true/predicted future positions; feature payload with overhead clearly separated. Raw plot data must be included. No plot may imply statistical uncertainty that was not estimated.

Software checks before experimental inference

Verify scene counts and exact 20/4 splits; paired physical targets; analytical target positions; constant-velocity error formula; ridge dual/primal agreement on a tiny example; unpenalized intercept; tie-to-orange color rule; aggregate RMSE versus pooled squared errors; zero-denominator handling; reproducible projection hash.

Verify no held-out row contributes to feature means/scales or reader fitting. Ensure target labels and metadata are never embedded into feature arrays. Modifying frames 61-150 must leave the actual tensor passed to the encoder unchanged; test that input-construction property without extra model runs.

Resume only if video hash, input-index list, code hash, config hash, source revision and checkpoint hash match. Do not reuse old embeddings from full-duration videos. Any cached artifact lacking matching provenance must be recomputed after approval.

Acceptance: all 24 clips accounted for; 24 finite full embeddings; 24 finite projected embeddings; exactly one out-of-fold prediction per clip per representation; no omitted groups; every decision reproducible from saved CSV and JSON.

12 / Worked checks, references and handoff

Hand-computable scene check (not an inference result)

For scene s=0, x0=-40 m and y0=140 m. At t=4 seconds the true x coordinate is -32 m. The true y coordinate is 61.6 m for g=9.8 and 100.8 m for g=4.9. Both colors have the same target. At t=5 seconds the baseline-gravity y coordinate is 17.5 m, safely above the lower boundary for a radius-five ball.

Interpretation checklist

Distinguish a narrow successful reader from a physics-engine claim. Predicting at two already-seen gravities does not show interpolation to unseen gravity. A linear projection does not identify semantic feature columns. Failure of a linear color reader does not establish irrecoverable color loss. Appearance changes induced by projection are distinct from original JEPA abstraction.

Source anchors

Official architecture and checkpoint source: https://github.com/facebookresearch/vjepa2/tree/204698b45b3712590f06245fbfba32d3be539812

Checkpoint: https://dl.fbaipublicfiles.com/vjepa2/vjepa2_1_vitb_dist_vitG_384.pt

Related probing study: Interpreting Physics in Video World Models, arXiv:2602.07050. This pilot is not a reproduction of that paper. Its dataset, ridge settings, projection and decision rules are explicit proposed choices for this small experiment.

Copyable handoff

Implement only the protocol in this package for the selected environment. First generate and validate the 24 colored-ball videos and present every clip for inspection. Stop before inference and wait for my explicit dataset approval. After approval, use the fixed first-two-second input, frozen final-layer JEPA features, seeded 16-D projection, six grouped folds and ridge calculations exactly as specified. Return all metrics, plots, artifacts and provenance, even for a null result. Do not expand the experiment or change its thresholds.

Current status: design documents prepared; no pilot videos generated, no checkpoint inference performed, no readers fitted. The user retains the decision about where to execute.